

EXERCISE 5: SEARCH FUNCTIONALITY



Student Management System

MVC Pattern with Jakarta EE & JSTL

Search

Show All

+ Add New Student

Search results for: **nguyên**

ID	Student Code	Full Name	Email	Major	Actions
6	IT123	Lê Quang Nguyễn	lequangnguyen@gmail.com	Computer Science	Edit Delete



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Search

Show All

+ Add New Student

Search results for: **SV005**

ID	Student Code	Full Name	Email	Major	Actions
5	SV005	David Wilson	david.w@email.com	Computer Science	Edit Delete



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Search

Show All

+ Add New Student

Search results for: **emily.j@email.com**

ID	Student Code	Full Name	Email	Major	Actions
2	SV002	Emily Johnson	emily.j@email.com	Information Technology	Edit Delete



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ID	Student Code	Full Name	Email	Major	Actions
				All Majors Filter	
6	IT123	Lê Quang Nguyễn	lequangnguyen@gmail.com	Computer Science	Edit Delete
5	SV005	David Wilson	david.w@email.com	Computer Science	Edit Delete
4	SV004	Sarah Davis	sarah.d@email.com	Data Science	Edit Delete
3	SV003	Michael Brown	michael.b@email.com	Software Engineering	Edit Delete
2	SV002	Emily Johnson	emily.j@email.com	Information Technology	Edit Delete
1	SV001	John Smith	john.smith@email.com	Computer Science	Edit Delete

In the DAO layer, a `searchStudents(String keyword)` method uses a `PreparedStatement` with SQL LIKE wildcards to search student code, full name and email, returning a `List<Student>` ordered by id descending. The controller reads the keyword parameter, handles empty input by falling back to the full list, calls the DAO search method, and forwards both students and keyword to `student-list.jsp`. The JSP view provides a GET search form with `action=search`, preserves the keyword in the input, shows a “Search results for: [keyword]” message, and conditionally displays a “Show All” button when a search is active.

EXERCISE 6: SERVER-SIDE VALIDATION

Edit Student

Student Code:

IT456

Full Name:

hiếu

Email (optional):

le

Major:



Vui lòng bao gồm '@' trong địa chỉ email. 'le' bị thiếu '@'.

Business Administration



Update



Cancel

EXERCISE 7: SORTING & FILTERING



Student Management System

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[Search](#)[+ Add New Student](#)

ID	Student Code	Full Name	Email	Major	Actions
				Computer Science	
6	IT123	Lê Quang Nguyễn	lequangnguyen@gmail.com	Computer Science	Edit Delete
5	SV005	David Wilson	david.w@email.com	Computer Science	Edit Delete
1	SV001	John Smith	john.smith@email.com	Computer Science	Edit Delete



Student Management System

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[Search](#)[+ Add New Student](#)

ID	Student Code	Full Name	Email	Major	Actions
				Information Technology	
2	SV002	Emily Johnson	emily.j@email.com	Information Technology	Edit Delete

In the DAO layer, we add methods to sort students by validated columns and order, and to filter students by major using a PreparedStatement, with an optional combined method that handles both. The controller reads sortBy, order, and major parameters, calls the corresponding DAO methods, and forwards the results plus state (selected major, sort column, sort direction) to the JSP. The student-list.jsp view provides clickable sortable column headers and a major dropdown filter, preserving the current sort/filter state and allowing users to clear the filter easily.